Executive Summary: Zepto Inventory & Pricing Analysis

Project Overview: A Strategic Review of E-commerce Catalog Health

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1. Objective

The goal of this analysis was to evaluate the "Catalog Health" of the Zepto inventory dataset. By utilizing advanced SQL querying, the project aimed to identify revenue leakage due to stock-outs, evaluate the competitiveness of current pricing strategies, and categorize inventory for optimized warehouse management.

2. Key Business Insights

📉 **Revenue at Risk (Stock-out Impact)**

The analysis identified a significant portion of high-margin inventory currently listed as **Out of Stock**.

- **Critical Finding:** The **Ice Cream & Desserts** category holds the highest potential revenue loss.
- **Recommendation:** Prioritize restocking cycles for "High-Velocity" SKUs (items with high demand but low stock) to prevent customers from switching to competitors.

💰 **Pricing & Discount Strategy**

We benchmarked category-level discounts against the platform-wide average of **7.62%**.

- **Observation:** Premium categories like **Home & Cleaning** are currently under-discounted compared to market standards, which may be affecting conversion rates.
- **Observation:** A "Price-per-Gram" analysis revealed that bulk-packaged items offer **0.3305 %** better value, suggesting a marketing opportunity to push "Value Packs" i.e 1.3823 % to price-sensitive customers.

📦 **Inventory Classification (ABC Analysis)**

Using the Pareto Principle (80/20 rule), the inventory was segmented into three priority tiers:

- **Tier A (High Value):** 20% of products contribute to ~70% of total inventory value. These require strict stock accuracy.
- **Tier B & C (Volume Drivers):** These items drive traffic but have lower individual margins.

3. Data Integrity & Cleaning

Before analysis, the raw dataset underwent a rigorous "Grooming" process:

- **Currency Normalization:** Converted raw scraped data from Paise to Rupees (INR) for standard financial reporting.
- **Anomaly Detection:** Identified and removed 0-priced "ghost listings" that would have skewed average revenue metrics.
- **Categorization:** Imputed missing values for uncategorized products to ensure 100% data visibility.

4. Technical Conclusion

This project demonstrates the transition from raw data annotation to **Business Intelligence**. By utilizing PostgreSQL Window Functions and CTEs, we transformed 10,000+ rows of static data into a dynamic roadmap for inventory optimization.